

OLA RIDE INSIGHTS

SUBMITTED TO:
LABMENTIX

SUBMITTED BY:
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Course:
DATA ANALYTICS

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MODULES

- MYSQL - For Data Cleaning, EDA and KPI
- POWERBI – For Building Dashboard
- STREAMLIT – lightweight interactive app

Introduction

The OLA project focuses on analyzing ride-hailing service data to uncover customer behavior, driver performance, ride cancellations, and payment trends. The dataset contains over 100,000 booking records with details such as booking status, reasons for incomplete rides, customer and driver ratings, ride distances, booking values, and payment methods. By examining this data, the project aims to provide actionable insights that can help improve service reliability, enhance customer experience, reduce cancellations, and optimize operational efficiency. This analysis is highly relevant for decision-making in the ride-hailing industry, where customer satisfaction and operational performance directly impact revenue.

Methodology

1. Data Collection & Preparation

- The raw booking dataset (OLA) was imported into MySQL for structured storage and query execution.
- Data cleaning was performed to handle missing values, standardize booking statuses, and ensure consistency across columns such as payment methods and cancellation reasons.

2. Exploratory Data Analysis (EDA)

- SQL queries were used to analyze booking trends, cancellation counts, payment preferences, and ride distances.
- Aggregations like total rides, cancellations by customer vs driver, and rating distributions were derived.

3. Business KPIs & DAX in Power BI

- KPIs such as total rides, completed rides, cancellation rate, and top customers by bookings were created.
- DAX measures were developed for calculating booking status counts, total cancellations, and identifying high-value customers.

4. Visualization & Dashboarding

- Power BI was used to design interactive dashboards.
- Visuals included cancellation rate breakdowns, top customers, preferred payment methods, and booking trends over time.
- Conditional formatting and filters were applied for better interpretation of customer and driver behaviors.

Key Findings

Ride Completion vs Cancellations

- A significant portion of rides were canceled, with notable differences between cancellations by customers and by drivers.
- Customer-driven cancellations were often higher than driver cancellations.

Customer Behavior

- A small group of customers booked disproportionately higher rides, showing loyalty and reliance on the service.
- Top 5 customers accounted for a noticeable share of total bookings.

Payment Method Insights

- Cash payments remained dominant in certain booking categories, though digital transactions were also prevalent.
- Specific payment modes were associated with higher booking values.

Ratings Distribution

- Both customer and driver ratings indicated trends in service satisfaction, highlighting areas for quality improvement.

Operational Gaps

- Incomplete rides and their reasons revealed inefficiencies such as driver unavailability or customer no-shows.
- These gaps directly affected service reliability and customer trust.

CODE:

MYSQL:

1.Retrieve all successful bookings

```
SELECT *  
FROM rides  
WHERE Booking_Status = 'Success';
```

2.Find the average ride distance for each vehicle type

```
SELECT Vehicle_Type, ROUND(AVG(Ride_Distance), 2) AS avg_distance  
FROM rides  
GROUP BY Vehicle_Type;
```

3.Get the total number of cancelled rides by customers

```
SELECT COUNT(*) AS `cancelled by customers`  
FROM rides  
WHERE Canceled_Rides_by_Customer IS NOT NULL;
```

4. List the top 5 customers who booked the highest number of rides

```
SELECT Customer_ID, COUNT(Booking_ID) AS total_rides  
FROM rides  
GROUP BY Customer_ID  
ORDER BY total_rides DESC  
LIMIT 5;
```

5.Get the number of rides cancelled by drivers due to personal and car-related issues

```
SELECT COUNT(*) AS total_personal_car_issues  
FROM rides  
WHERE Canceled_Rides_by_Driver = 'Personal & Car related issue';
```

6. Find the maximum and minimum driver ratings for Prime Sedan bookings

```
SELECT MAX(Driver_Ratings) AS max_rating, MIN(Driver_Ratings) AS  
min_rating  
FROM rides  
WHERE Vehicle_Type = 'Prime Sedan';
```

```
select * from rides;
```

7. Retrieve all rides where payment was made using UPI

```
SELECT *  
FROM rides  
WHERE Payment_Method = 'UPI';
```

8. Find the average customer rating per vehicle type

```
SELECT Vehicle_Type, ROUND(AVG(Customer_Rating),2) AS  
avg_customer_rating  
FROM rides  
GROUP BY Vehicle_Type;
```

9. Calculate the total booking value of rides completed successfully

```
SELECT SUM(Booking_Value) AS total_booking_value  
FROM rides  
WHERE Booking_Status = 'Success';
```

10. List all incomplete rides along with the reason:

```
SELECT Booking_ID, Booking_Status, Incomplete_Rides_Reason  
FROM rides  
WHERE Incomplete_Rides_Reason IS NOT NULL  
AND Incomplete_Rides_Reason != '';
```

Explanation:

1. Retrieve all successful bookings

- This query filters and displays all rides where the booking was completed successfully.

2. Average ride distance for each vehicle type

- Calculates the mean ride distance for every vehicle category to compare performance.

3. Total number of cancelled rides by customers

- Counts all rides where customers initiated the cancellation.

4. Top 5 customers with highest bookings

- Identifies the most frequent riders by counting their total bookings.

5. Rides cancelled by drivers due to personal/car issues

- Counts cancellations made by drivers citing personal or vehicle-related reasons.

6. Max & Min driver ratings for Prime Sedan

- Retrieves the highest and lowest driver ratings specifically for Prime Sedan rides.

7. All rides with UPI payment

- Filters rides where the payment method used was UPI.

8. Average customer rating per vehicle type

- Computes the mean customer rating across different vehicle categories.

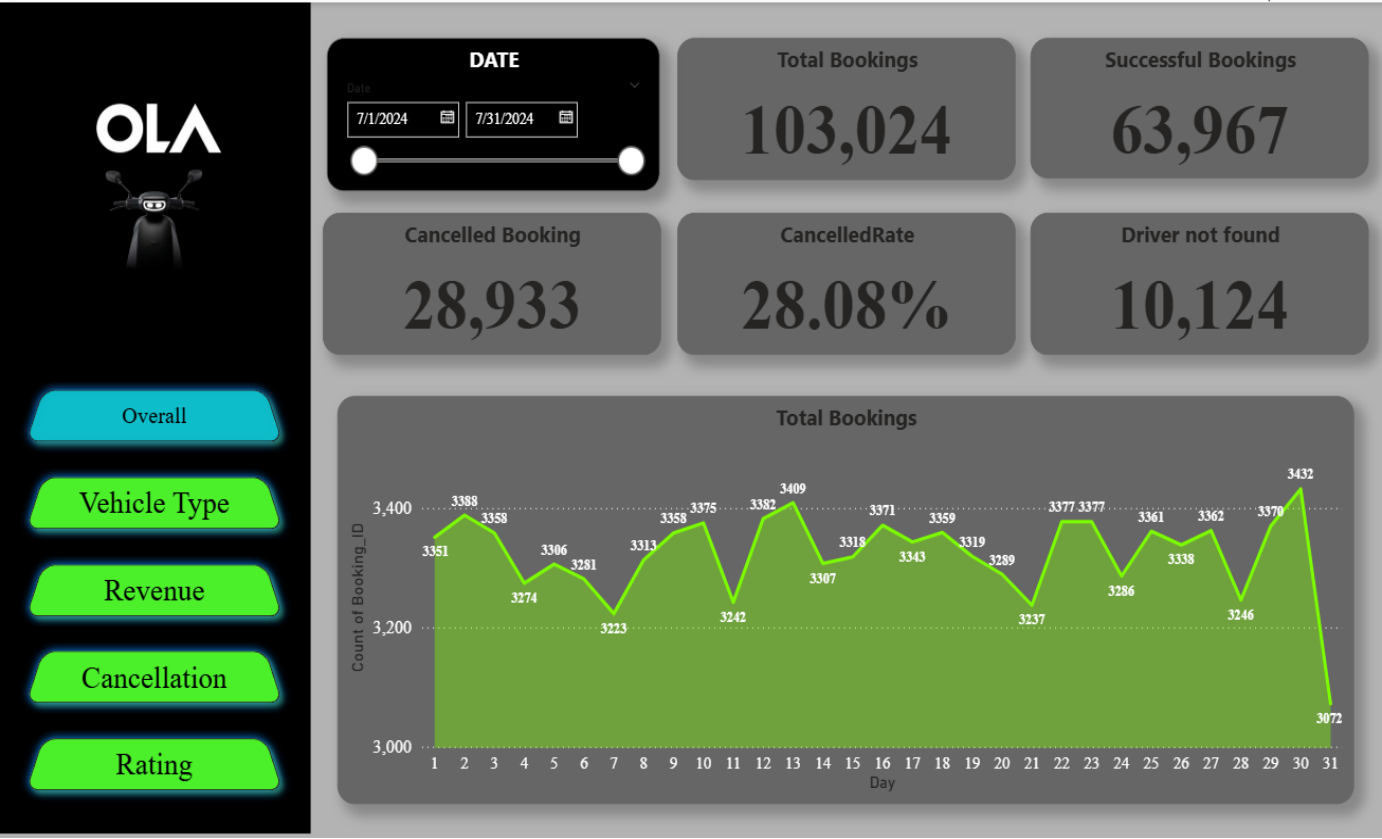
9. Total booking value of successful rides

- Adds up the booking values for all rides that were successfully completed.

10. List all incomplete rides with reasons

- Displays incomplete rides along with their corresponding failure reasons.

Screenshot:



OLA

Overall

Vehicle Type

Revenue

Cancellation

Rating

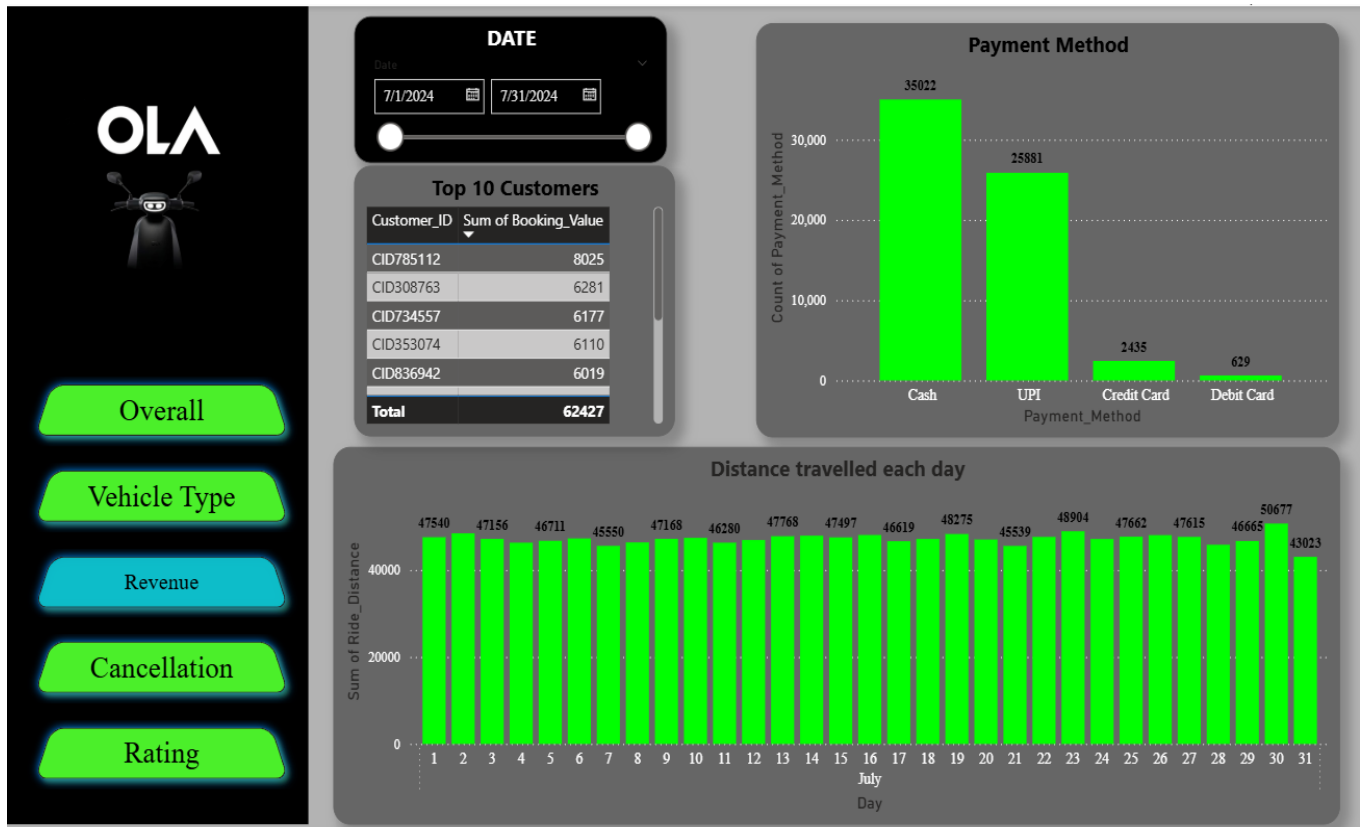
DATE

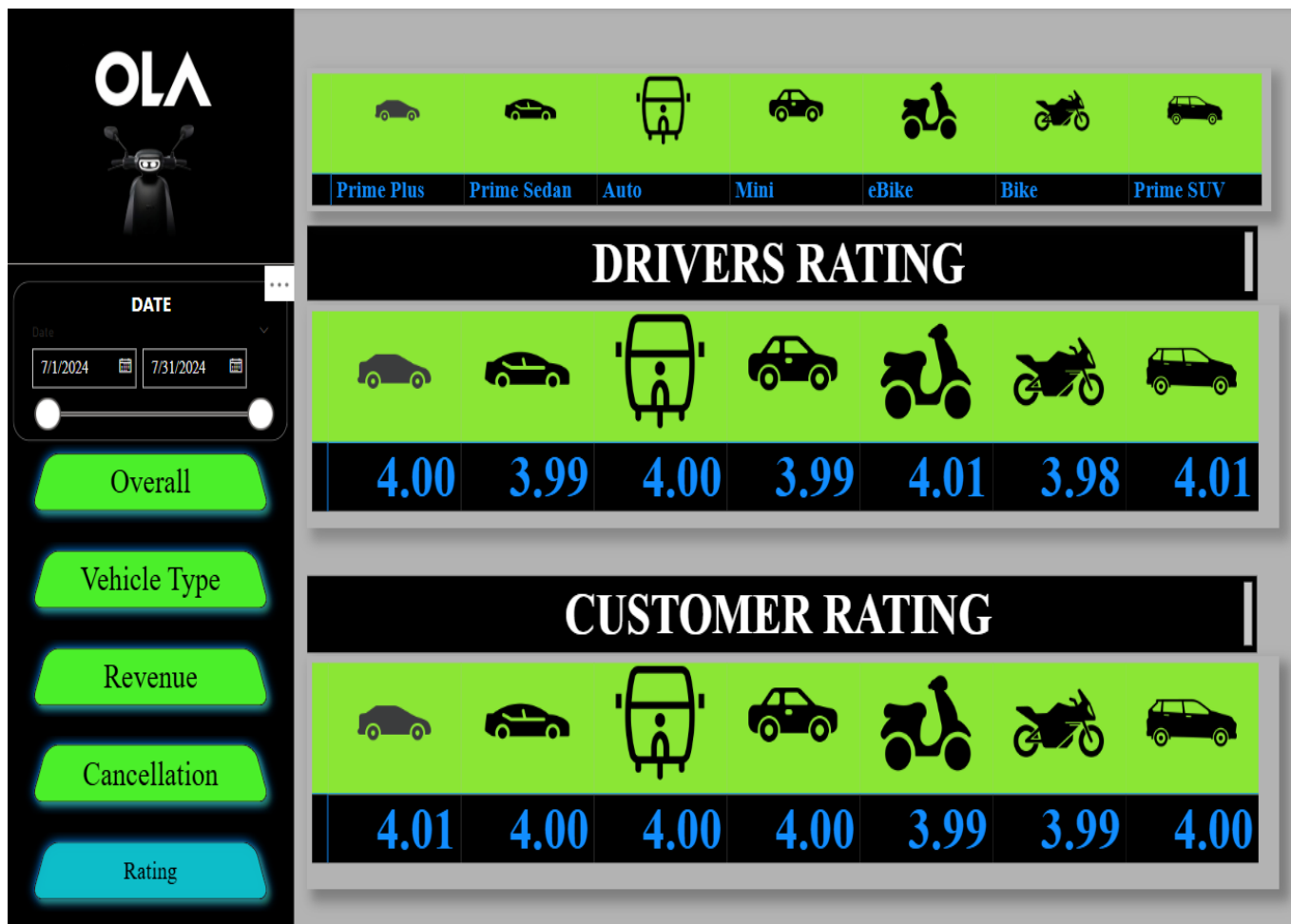
Date

7/1/2024

7/31/2024

Vehicle Images	Count of Booking_ID	Successful Rides	Total Ride Distance	Average Ride Distance	Total Amt Cancellations	Cancellation Rate	
	14877	9379	234,535	15.76	4061	3.94%	
	14816	9180	230,842	15.58	4206	4.08%	
	14755	9167	92,043	6.24	4178	4.06%	
	14707	9075	227,186	15.45	4157	4.03%	
	14662	9134	227,746	15.53	4096	3.98%	
	14655	8996	223,848	15.27	4146	4.02%	
	14552	9036	225,703	15.51	4080	3.97%	
Total	103024	63967	1,461,903	14.19	28933	28.08%	





Explanation

Page 1 – Ride Overview

- Total Rides / Completed Rides KPI – Shows the overall booking volume and success rate, highlighting OLA’s operational scale.
- Cancellation Breakdown (Customer vs Driver) – Visualizes cancellation patterns, showing that customer-driven cancellations are higher than driver ones.
- Trend of Rides Over Time – Tracks demand fluctuations, peak times, and seasonal variations.

Page 2 – Customer Insights

- Top 5 Customers by Bookings – Highlights loyal customers who generate the highest booking volume.
- Customer Ratings Distribution – Shows satisfaction levels, identifying trends in service experience.

- Customer vs Booking Value Chart – Links ride frequency with spending, showing high-value customers.

Page 3 – Driver & Vehicle Analysis

- Driver Ratings Distribution – Reflects service quality and areas for improvement.
- Cancellations by Drivers (Reasons) – Breaks down issues like personal or car-related problems.
- Average Ride Distance by Vehicle Type – Compares vehicle efficiency and suitability for different ride patterns.

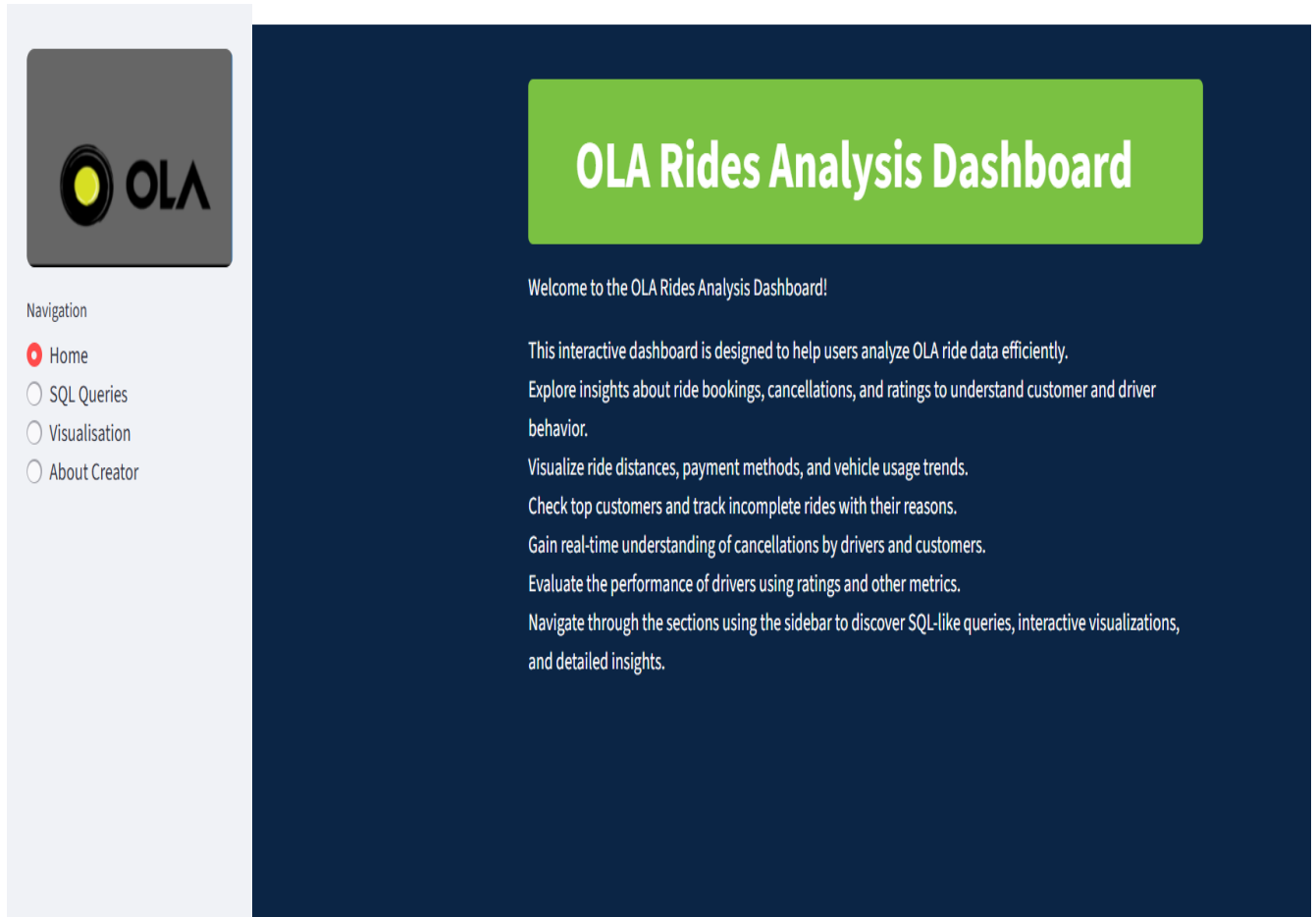
Page 4 – Payment & Revenue

- Payment Mode Distribution – Compares cash vs digital (UPI, card, wallet), showing customer preference.
- Revenue by Payment Method – Identifies which payment modes generate higher booking values.
- Total Booking Value KPI – Sums all successful ride earnings, a key profitability metric.

Page 5 – Operational Gaps

- Incomplete Rides with Reasons – Displays why rides failed (driver unavailability, customer no-show).
- Customer vs Driver Cancellation Share – Reinforces reliability issues impacting trust.
- Operational Efficiency KPI – Helps track how cancellations and incomplete rides affect service quality.

STREAMLIT OUTPUT:



Choose a Query

1. Retrieve all successful bookings

Generate

Download Search Refresh

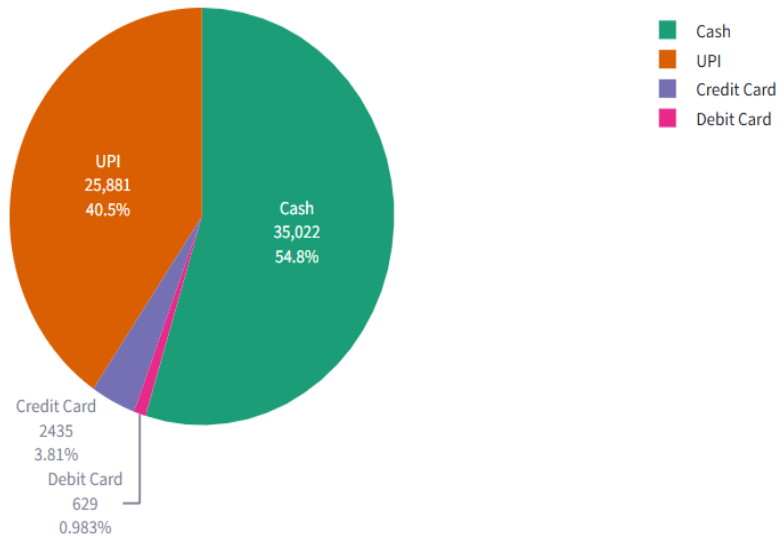
	Date	Time	Booking_ID	Booking_Status	Customer_ID	Vehicle_Type	Pickup_Location
1	2024-07-25 22:20:00	22:20:00	CNR2940424040	Success	CID225428	Bike	Magan
2	2024-07-30 19:59:00	19:59:00	CNR2982357879	Success	CID270156	Prime SUV	Sahak
4	2024-07-02 9:02:00	9:02:00	CNR1797421769	Success	CID939555	Mini	Rajaji
5	2024-07-13 4:42:00	4:42:00	CNR8787177882	Success	CID802429	Mini	Kadu
6	2024-07-23 9:51:00	9:51:00	CNR3612067560	Success	CID476071	Bike	Tumk
10	2024-07-29 23:33:00	23:33:00	CNR4787583516	Success	CID923404	Prime Plus	Hosu
11	2024-07-26 4:03:00	4:03:00	CNR7943634301	Success	CID647026	Prime Plus	Kamr
12	2024-07-27 13:18:00	13:18:00	CNR4524472111	Success	CID540929	Auto	Cox T
14	2024-07-16 9:54:00	9:54:00	CNR8181602032	Success	CID167642	Bike	Indira
15	2024-07-02 10:25:00	10:25:00	CNR8090918544	Success	CID640151	Bike	Maga

Download Results

All

Generate

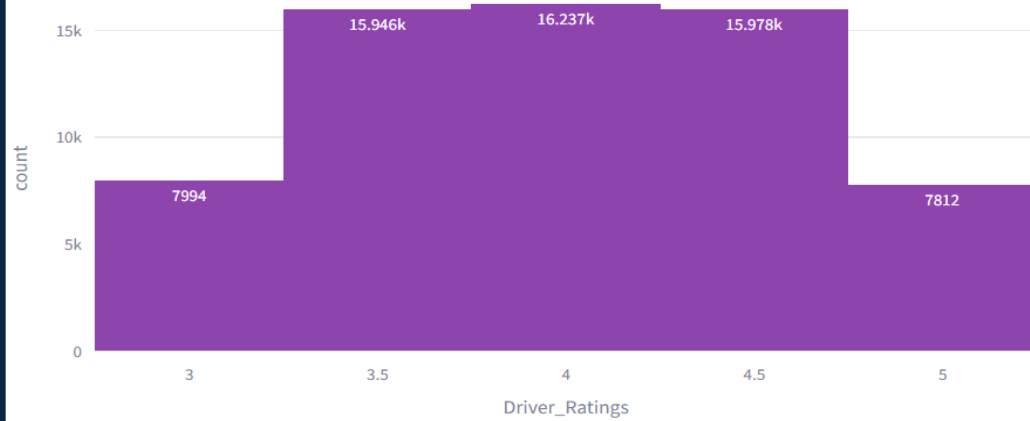
Payment Method Distribution

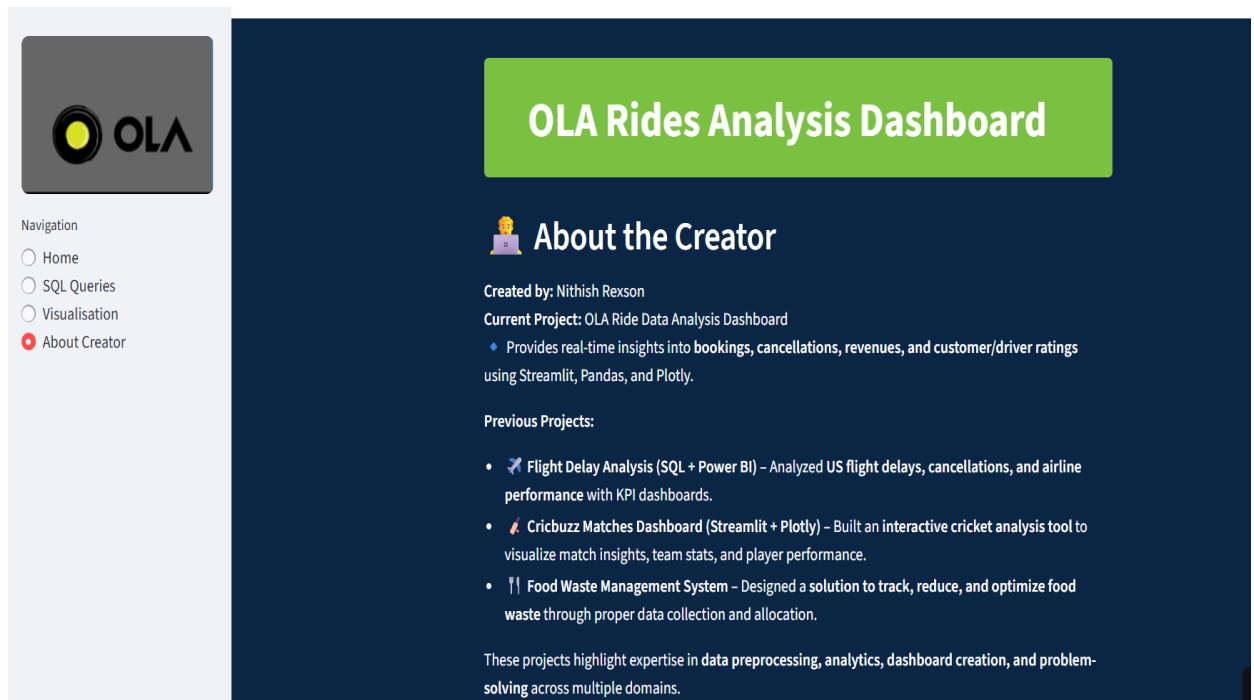


Driver Ratings

Generate

Driver Ratings Distribution





Explanation

SQL Queries Page Explanations

1. Retrieve all successful bookings – Shows all rides with booking status marked as “Success.”
2. Average ride distance per vehicle type – Displays mean ride distance for each vehicle type.
3. Total cancelled rides by customers – Counts all rides cancelled by customers.
4. Top 5 customers with highest rides – Lists the 5 customers who booked the most rides.
5. Rides cancelled by drivers due to Personal & Car issues – Counts rides cancelled for personal/car reasons.
6. Max & Min driver ratings for Prime Sedan – Shows highest and lowest driver ratings for Prime Sedan rides.
7. Rides with UPI payment – Lists all rides paid via UPI.
8. Average customer rating per vehicle type – Shows average customer rating for each vehicle type.
9. Total booking value of successful rides – Calculates total revenue from successful bookings.
10. All incomplete rides with reason – Displays rides that were incomplete along with the reason.

Visualisation Page Explanations

Revenue

- **Total Revenue** – Shows the sum of all booking values.
- **Revenue by Vehicle Type** – Bar chart of revenue for each vehicle type.
- **Revenue by Payment Method** – Pie chart of revenue split by payment method.

Cancellations

- **Driver Cancellations** – Total rides cancelled by drivers.
- **Customer Cancellations** – Total rides cancelled by customers.
- **All Cancellations** – Pie chart comparing driver vs customer cancellations.

Rating

- **Driver Ratings** – Histogram showing distribution of driver ratings.
- **Customer Ratings** – Histogram showing distribution of customer ratings.
- **Average Rating by Vehicle Type** – Bar chart of average customer rating per vehicle type.

Vehicle Type

- **Booking Status Trend** – Line or count showing status distribution for selected vehicle type.

Payment Method

- **Payments by Type** – Count of rides per payment method, or pie chart if “All” is selected.

CONCLUSION

The OLA Rides Analysis Dashboard provides a comprehensive view of ride operations, highlighting key trends in bookings, cancellations, revenue, and ratings. Successful bookings indicate customer demand patterns, while cancellations by drivers and customers reveal operational bottlenecks. Revenue analysis by vehicle type and payment method helps identify the most profitable segments and preferred payment channels. Customer and driver ratings offer insights into service quality and satisfaction levels. The identification of top customers allows targeted engagement strategies. Visualizations enable quick understanding of trends, such as ride distribution, revenue contributions, and cancellation patterns. Incomplete rides and their reasons highlight areas for service improvement. Overall, this dashboard equips stakeholders with actionable insights for optimizing operations, improving customer experience, and enhancing revenue performance. It serves as a powerful tool for data-driven decision-making in ride-hailing services.